Project: Worked examples
Title: Material strength margin
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

The factors of safety of a material against an applied (equivalent) stress on its yield and, when given, its ultimate strength, against a required factor; the strengths can be filled from a released material record so the margin cites its source.
$n_y = f_y/\sigma$; $n_u = f_u/\sigma$. The governing factor is the lower of n_y and n_u/k_u, where k_u (≥ 1) scales the ultimate margin to the yield basis; $u = n_{req}/n_{gov}$.
Ref: Shigley's Mechanical Engineering Design §1-10 — Design factor and factor of safety

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
sigma	Applied (equivalent) stress	120	MPa	spec: LOAD-SCHEDULE-01	0.01 … 5000 MPa
f_y	Yield strength	300	MPa	drawing: DRG-BR-701 rev A	1 … 5000 MPa
f_u	Ultimate tensile strength (optional)	450	MPa	drawing: DRG-BR-701 rev A	1 … 5000 MPa
k_u	Ratio of the ultimate to the yield safety factor required	1.5		spec: SPEC-MECH-03 design criteria	1 … 3
n_req	Required factor of safety on yield	1.5		spec: SPEC-MECH-03 design criteria	1 … 10

3. Assumptions

A1. Static loading; the applied stress is the equivalent stress at the critical point. (Fatigue and stability are separate checks.)

4. Working

[5] $n_y = f_y / \sigma$ Shigley §1-10
$= 300\ \text{MPa} / 120\ \text{MPa}$
$= 2.5$
Factor of safety on yield
[7] $n_u = f_u / \sigma$
$= 450\ \text{MPa} / 120\ \text{MPa}$
$= 3.75$
Factor of safety on ultimate strength
[8] $n_{gov} = \min(n_y, n_u / k_u)$
$= \min(2.5, 3.75 / 1.5)$
$= 2.5$
Governing factor (ultimate margin scaled to the yield basis)
[9] $u = n_{req} / n_{gov}$
$= 1.5 / 2.5$
$= 0.6$
Required over governing factor of safety

5. Results

SYMBOL	VALUE	DESCRIPTION
n_u	3.75	Factor on ultimate
n_y	2.5	Factor on yield
n_gov	2.5	Governing factor
u	0.6	Utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Factor of safety against the required factor	u = 0.6	u ≤ 1	0.6	Shigley §1-10

7. Applicability

No warnings.

8. Verification

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